University of Maryland—full professor at age twenty-nine, and with no more college education than a bachelor's degree.

While teaching electrical engineering at Maryland, Weber prepared for a career change: He worked toward, and completed, a Ph.D. in physics at Catholic University, in part under the same person as had been John Wheeler's Ph.D. adviser, Karl Herzfeld. From Herzfeld, Weber learned enough about the physics of atoms, molecules, and radiation to invent, in 1951, one version of the mechanism by which lasers work, but he did not have the resources to demonstrate his concept experimentally. While Weber was publishing his concept, two other research groups, one at Columbia University led by Charles Townes and the other in Moscow led by Nikolai Gennadievich Basov and Aleksandr Michailovich Prokharov, independently invented alternative versions of the mechanism, and then went on to construct working lasers.[2] Though Weber's paper had been the first publication on the mechanism, he received hardly any credit; the Nobel Prize and patents went to the Columbia and Moscow scientists. Disappointed, but maintaining close friendships with Townes and Basov, Weber sought a new research direction.

As part of his search, Weber spent a year in John Wheeler's group, became an expert on general relativity, and with Wheeler did theoretical research on general relativity's predictions of the properties of gravitational waves. By 1957, he had found his new direction. He would embark on the world's first effort to build apparatus for detecting and monitoring gravitational waves.

Through late 1957, all of 1958, and early 1959, Weber struggled to invent every scheme he could for detecting gravitational waves. This was a pen, paper, and brainpower exercise, not experimental. He filled four 300-page notebooks with ideas, possible detector designs, and calculations of the expected performance of each design. One idea after another he cast aside as not promising. One design after another failed to give high sensitivity. But a few held promise; and of them, Weber ultimately chose a cylindrical aluminum bar about 2 meters long, a half meter in diameter, and a ton in weight, oriented broadside to the incoming waves (Figure 10.4 below).

As the waves' tidal force oscillates, it should first compress, then stretch, then compress such a bar's ends. The bar has a natural mode of vibration which can respond resonantly to this oscillating tidal force, a